AeroGuard: Comparative Analysis of Regression and Ensemble Learning Algorithms for Predicting Aircraft Engine Remaining Useful Life

1. Aim

The goal of this study is to build and compare various regression models to predict the Remaining Useful Life (RUL) of aircraft engines using the data provided by the NASA C-MAPSS Turbofan Engine Degradation Simulation Dataset. This study will explore how various machine learning techniques can be used to predict the degradation of aircraft engines and how these predictions can help improve maintenance planning.

2. Problem Statement

Aircraft engines are subject to gradual degradation. If this degradation isn't detected then it can cause unexpected failures, higher maintenance costs and safety issues. The ability to predict how long an engine will last is therefore of interest so that maintenance can be planned. This project aims to explore different regression and ensemble learning algorithms in order to predict the remaining useful life of aircraft engines and compare them to find suitable models.

3. Objectives

The main purpose of this study is to implement various regression models to predict the remaining useful life of an aircraft engine. The models will be compared based on their evaluation metrics to determine which regression model produces the most accurate results. An understanding of how each regression model and ensemble learning techniques work will be gained.

4. Literature Review

The area of predictive maintenance is an important area of research within the aviation industry. The unexpected failure of aircraft engines can cause issues with increased maintenance costs, disruption and safety. Remaining Useful Life (RUL) estimation predicts how many operation cycles will occur before the end of an aircraft engine's Useful Life. Data driven methods use engine operating conditions and sensor data to identify degradation patterns and predict RUL.

The NASA Turbofan Engine Degradation Simulation dataset was presented by Saxena and Goebel [1]. This dataset is widely known as the C-MAPSS dataset, and contains simulated degradation patterns for aircraft engines which can be used to develop and test prognostic algorithms. The dataset has been useful to compare RUL estimation methods across various operating conditions.

A deep convolutional neural network was proposed by Babu et al. to estimate RUL [2]. This research displayed the potential of automatic feature extraction methods. Zhao et al. looked at learning degradation patterns for RUL estimation of aircraft engines [3]. These studies demonstrate how data driven methods can be used to help with predictive maintenance by using sensor data.

Recent research has focused on developing hybrid deep learning methods. Patra et al. looked at using a hybrid CNN LSTM GRU model to estimate RUL for aircraft engines [4]. Research published in the IEEE Sensors Journal in 2025 used a graph neural network model for RUL prediction in turbofan engines [5]. However, the performance of these advanced methods depend heavily on the data set, pre-processing and evaluation.

Traditional regression and ensemble learning algorithms can also be used for RUL prediction. Linear regression can be used as it is easily interpretable. Ridge and Lasso can be used with regularisation. Support Vector Regression can model non-linear relationships. Ensemble methods such as Bagging, AdaBoost, Gradient Boosting and XGBoost can be used. Comparison of these algorithms can be carried out to determine which performs better.

Based on the research, this study compares ten regression and ensemble learning algorithms. The models have been tested using Mean Absolute Error, Root Mean Squared Error and the coefficient of determination. This study uses a consistent pre-processing and evaluation method to find which model performs best. This will look to see how well traditional regression and ensemble methods can perform for RUL prediction in aircraft engines using the NASA C-MAPSS FD004 dataset.

5. Methodology

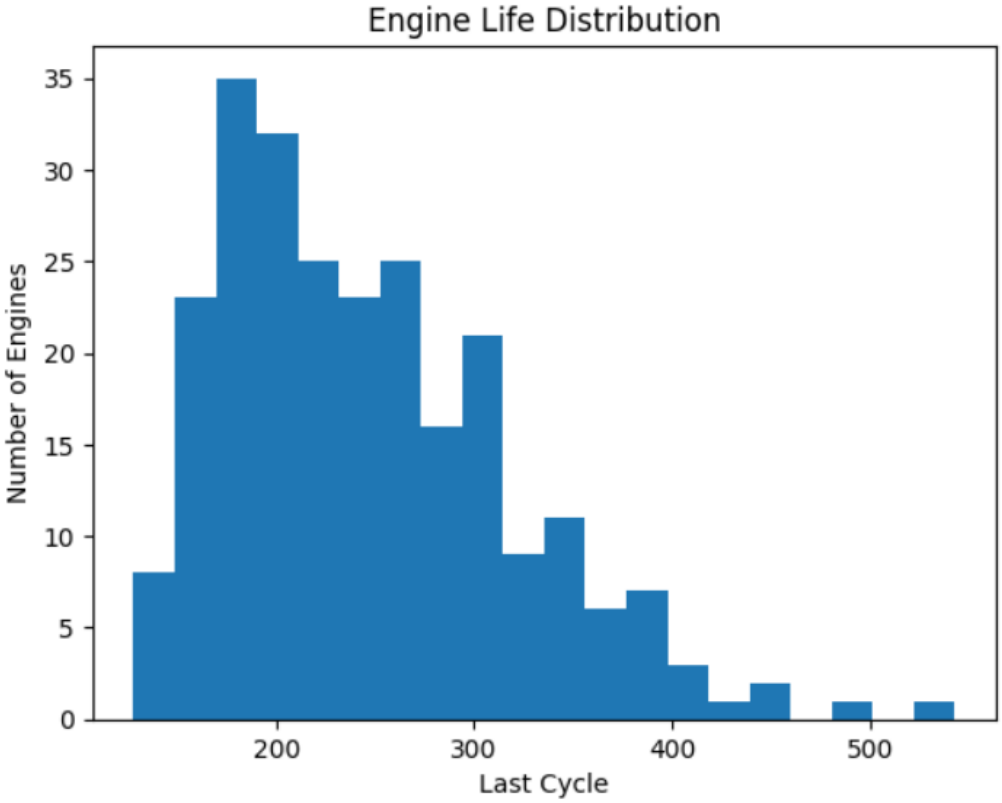
The NASA C-MAPSS FD004 dataset will be used to provide simulated operational conditions and sensor readings for aircraft engines. An initial analysis of the dataset will be undertaken to help understand the structure of the dataset, identify any missing/irrelevant values and understand the behaviour of the sensors. The remaining useful life (RUL) value for each training observation will be calculated by taking the difference between the last training cycle recorded for the engine and the current cycle. Feature selection will take place before applying suitable pre-processing techniques. The data will be split into training and validation sets based on engine ID's to prevent data leakage. A number of different models will be implemented and compared. These include: Simple Linear Regression, Multiple Linear Regression, Ridge Regression, Lasso Regression, Generalized Linear Model (GLM), Support Vector Regression (SVR), Bagging Regressor, AdaBoost Regressor, Gradient Boosting Regressor and XGBoost Regressor. The models will be assessed using mean absolute error (MAE), root mean squared error (RMSE) and coefficient of determination (R2). The results will be presented using tables and visualisations.

5.1 Data Description and Pre-processing

The dataset used in this study is the NASA C-MAPSS FD004 turbofan engine degradation dataset. The dataset consists of engine ID's, cycles, 3 operating settings and 21 sensor readings. There are 61,249 observations in the training data set from 249 engines and 41,214 observations in the testing data set from 248 engines. The test RUL labels are provided separately.

An initial data analysis showed the training dataset had no missing values. Summary statistics were produced to observe the distribution and characteristics

of the numerical variables.



5.2 Exploratory Data Analysis

An exploratory data analysis was carried out to gain insight into the distribution of engine life-cycles and the sensor data. A histogram was created to show the distribution of final cycles of training engines. Also a plot was created to show how the sensors varied throughout different operating cycles within one engine.

5.3 RUL Target and Feature Selection

For each training example the RUL was calculated by taking the difference between the last training cycle and the current cycle. As there are often very high initial life-periods, the target was restricted to a maximum value of 125.

The engine ID and cycle were dropped from the features. Any constant features were also dropped. Leaving us with 24 features to work with.

5.4 Train and Test Split

The data was split into training and test sets using GroupShuffleSplit, and the group was based on the engine ID. This resulted in 49294 training examples from 199 engines and 11955 test examples from 50 engines. This ensures that no examples from the same engine will show up in both training and testing.

StandardScaler was applied to the training set within pipelines. This means the scaling parameters can be learnt from the training data, not the testing set.

6. Implementation Tools

This project will be implemented in Python using Jupyter Notebook. The data will be manipulated using libraries such as Pandas, NumPy, visualised using Matplotlib and Seaborn, regression models will be implemented and evaluated using Scikit-learn and predictions will be made using gradient boosting from XGBoost. The data will come from the NASA C-MAPSS dataset.

6.1 Model Implementation

The regression models were implemented using Scikit-learn and XGBoost. The models tested were Simple Linear Regression, Multiple Linear Regression, Ridge Regression, Lasso Regression, Gaussian Generalised Linear Model, Support Vector Regression, Bagging Regressor, AdaBoost Regressor, Gradient Boosting Regressor and XGBoost Regressor.

The performance of each model was analysed using Mean Absolute Error, Root Mean Squared Error and R squared metric on the same holdout test set. Bar charts were created to compare the differences in MAE and RMSE of each model. Actual vs Predicted and Residual plots were created to analyse the predictions of the best model on the validation set.

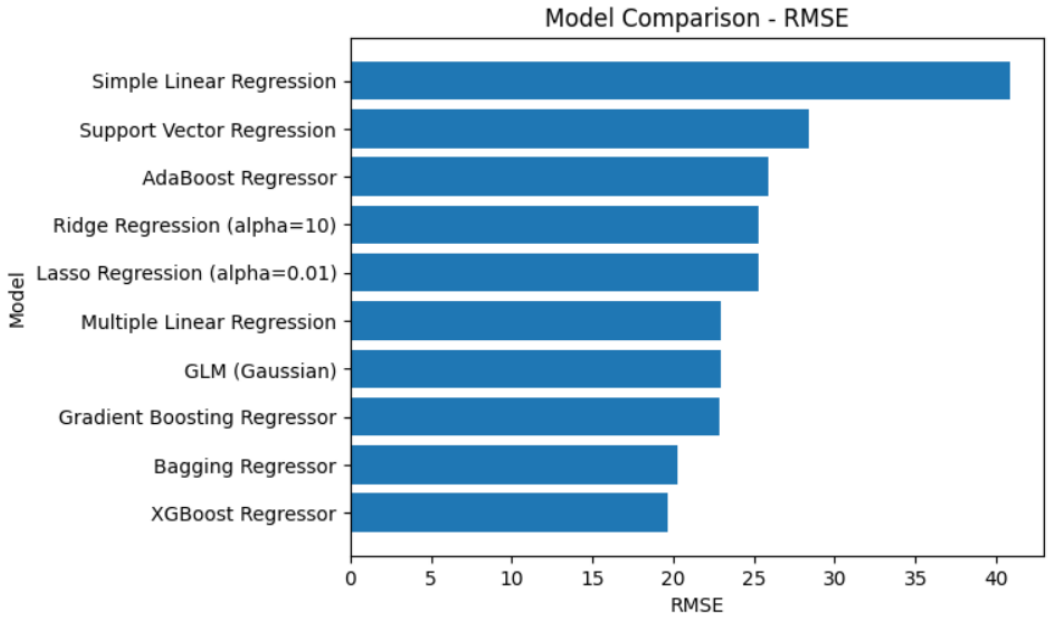
7. Results and Analysis

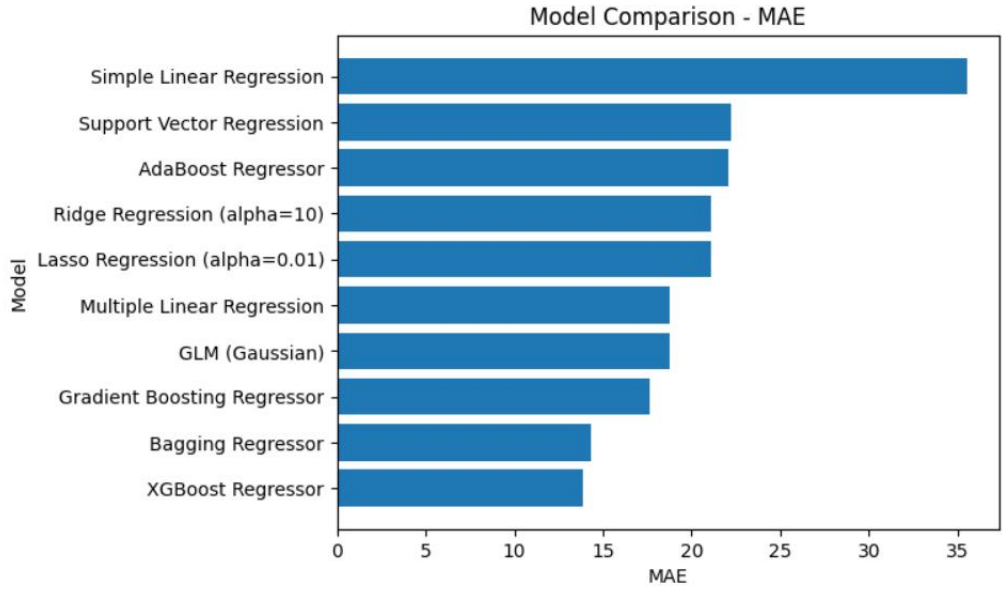
The 10 regression algorithms have been compared using MAE, RMSE and R squared metrics on the same validation set of 50 holdout engines. Lower values for MAE and RMSE suggest the model has less error in its predictions. Higher values for R squared suggest it performs better in explaining the data when compared to simply predicting the mean.

7.1 Validation Results

Below is a table showing the validation results:

Regression Model	**MAE**	**RMSE**	**R^2**
Simple Linear Regression	35.5791	40.8925	-0.0008
Multiple Linear Regression	18.7594	22.9666	0.6843
Ridge Regression	21.0916	25.2811	0.6175
Lasso Regression	21.0741	25.2764	0.6176
Gaussian GLM	18.7594	22.9666	0.6843
Support Vector Regression	22.2510	28.4093	0.5170
Bagging Regressor	14.2874	20.2242	0.7552
AdaBoost Regressor	22.0856	25.8605	0.5998
Gradient Boosting Regressor	17.5950	22.8195	0.6883
XGBoost Regressor	13.8323	19.6484	0.7689





7.2 Comparative Analysis

XGBoost was the overall best performer with an MAE of 13.8323, RMSE of 19.6484 cycles and R2 value of 0.7689. Second came Bagging Regressor with an MAE of 14.2874, RMSE of 20.2242 cycles and R2 value of 0.7552.

Gradient Boosting also had a good performance with a RMSE of 22.8195 cycles. Multiple Linear Regression and Gaussian GLM had the same performance with a RMSE of 22.9666 cycles and R2 value of 0.6843.

Simple Linear Regression using only sensor_2 performed very poorly with an R2 of -0.0008, meaning it performed slightly worse than just predicting the average in the validation set. This shows that just measuring one sensor was not enough to give a good representation of the variation in RUL.

Overall the validation set shows that the ensemble tree-based methods performed better than the linear and support vector models, especially XGBoost and Bagging.

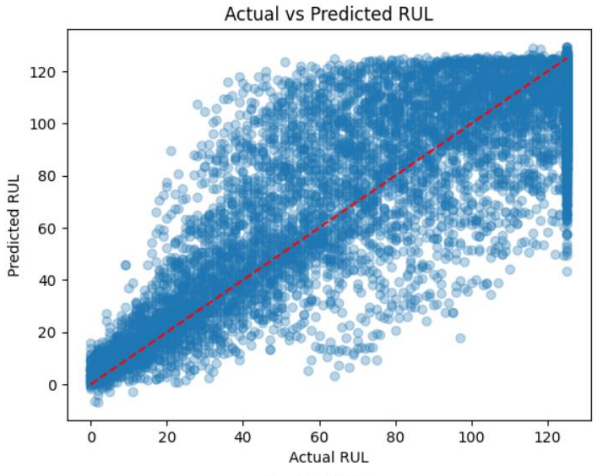
7.3 Official Test Set Evaluation

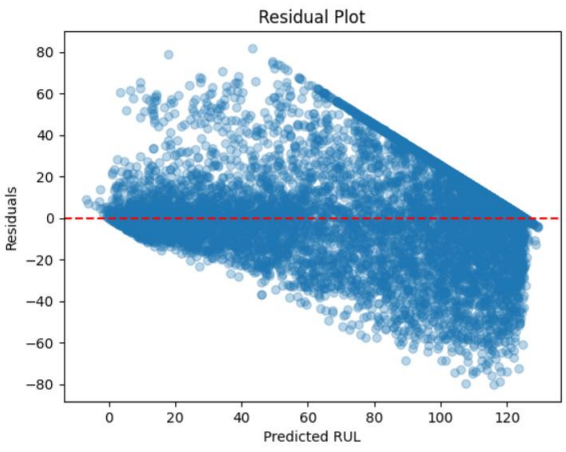
The notebook also evaluates XGBoost on the official FD004 test set. The results are shown below.

Metric	Test Result
MAE	22.650 cycles
RMSE	30.337 cycles
R^2	0.6904

The test-set MAE indicates an average absolute prediction error of approximately 22.65 cycles. The RMSE of 30.337 cycles reflects the additional influence of larger prediction errors. The R^2 value of 0.6904 indicates that the predictions explain approximately 69.04% of the variation in the supplied test RUL labels relative to a mean-prediction baseline.

The test results are not directly interchangeable with the validation results because the training target is capped at 125 cycles, whereas the supplied test RUL labels are not capped in the notebook. This difference should be considered when interpreting the evaluation.






```
Best model: XGBoost Regressor
```

7.4 Visual Analysis

We have created the following plots:

- Histogram of Engine Life.
- Sensor data vs. Cycles for a single engine.
- Distribution of RUL targets for Train and Validation.
- Bar plots to compare errors of different models (RMSE & MAE).
- RUL values Actual vs Predicted.
- Residual plot showing the results of validation model that performed best.

These plots help to identify trends in the engine lives, compare errors between models and find any discrepancies between actual and predicted RUL values. The residual plot will show us any systematic error in the predictions.

8. Conclusion

This study has implemented and compared 10 Regression and Ensemble learning models for predicting the remaining useful life of aircraft engines using the NASA C-MAPSS FD004 dataset. We have preprocessed our data, generated RUL targets, performed engine specific validation and evaluated our models. We found that our XGBoost model performed best with the lowest MAE, RMSE and highest R squared value of 13.8323, 19.6484 and 0.7689 respectively. Our bagging regressor also performed well. Our official test set evaluation showed that our XGBoost model had a MAE, RMSE and R squared value of 22.650, 30.337 and 0.6904. This shows that using Regression and Ensemble learning methods can be useful for predicting the remaining useful life of aircraft engines. However, there is a discrepancy in the performance of our model when validating versus testing, as well as the fact that RUL targets are capped. This shows how important it is to ensure that we evaluate our models consistently, before deploying them. This study will provide a basis for comparing models in future studies looking into predictive maintenance.

9. Future Scope

We can improve this project by including time series features, trying more complex deep learning models and predicting the uncertainty in our RUL values. Future work could look into how sensor failure, varying conditions and real engine data affects our predictions. This would make our system more practical for a real world predictive maintenance system.

10. References

[1] G. S. Babu, P. Zhao, and X.-L. Li, “Deep convolutional neural network based regression approach for estimation of remaining useful life,” in *Database Systems for Advanced Applications*, 2016, pp. 214–228, doi: 10.1007/978-3-319-32025-0_14.

[2] Z. Zhao, B. Liang, X. Wang, and W. Lu, “Remaining useful life prediction of aircraft engine based on degradation pattern learning,” *Reliability Engineering & System Safety*, vol. 164, pp. 74–83, Aug. 2017, doi: 10.1016/j.ress.2017.02.007.

[3] X. Li, Q. Ding, and J.-Q. Sun, “Remaining useful life estimation in prognostics using deep convolution neural networks,” *Reliability Engineering & System Safety*, vol. 172, pp. 1–11, Apr. 2018, doi: 10.1016/j.ress.2017.11.021.

[4] K. C. Patra, R. Sethi, and D. K. Behera, “Estimation of the remaining useful life of aircraft engines using a CNN-LSTM-GRU hybrid model,” *International Journal of System Assurance Engineering and Management*, vol. 16, no. 12, pp. 3968–3982, 2025, doi: 10.1007/s13198-025-02911-4.

[5] X. Cao, B. Cheng, X. Yang, Y. Duan, and J. Gao, “Remaining useful life prediction method for turbofan engines based on graph neural networks,” *IEEE Sensors Journal*, vol. 25, no. 8, pp. 13566–13578, Apr. 2025, doi: 10.1109/JSEN.2025.3546498.

[6] A. Saxena and K. Goebel, “Turbofan engine degradation simulation data set,” NASA Ames Prognostics Data Repository, NASA Ames Research Center, Moffett Field, CA, USA, 2008. [Online].